



MATHEMATICS METHODS

Unit 1 and Unit 2

Formula Sheet

(For use with Year 11 examinations and response tasks)

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This document is valid for teaching and examining from 1 July 2015.

Measurement

Circle: $C = 2\pi r = \pi D$, where C is the circumference,
 r is the radius and D is the diameter
 $A = \pi r^2$, where A is the area

Triangle: $A = \frac{1}{2}bh$, where b is the base and h is the perpendicular height

Parallelogram: $A = bh$

Trapezium: $A = \frac{1}{2}(a + b)h$, where a and b are the lengths of the parallel sides

Prism: $V = Ah$, where V is the volume and A is the area of the base

Pyramid: $V = \frac{1}{3}Ah$

Cylinder: $S = 2\pi rh + 2\pi r^2$, where S is the total surface area
 $V = \pi r^2h$

Cone: $S = \pi rs + \pi r^2$, where s is the slant height
 $V = \frac{1}{3}\pi r^2h$

Sphere: $S = 4\pi r^2$
 $V = \frac{4}{3}\pi r^3$

Functions and graphs

Lines and Linear relationships

For points $P(x_1, y_1)$ and $Q(x_2, y_2)$

Mid-point of P and Q :

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Gradient of the line through P and Q :

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Equation of the line through P with slope m : $y - y_1 = m(x - x_1)$

Parallel lines:

$$m_1 = m_2$$

Perpendicular lines:

$$m_1 m_2 = -1$$

General equation of a line:

$$ax + by + c = 0 \text{ or } y = mx + c$$

Quadratic relationships

For the general quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$

Completing the square: $ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + \left(c - \frac{b^2}{4a} \right)$

Discriminant: $\Delta = b^2 - 4ac$

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Graphs and Relations

Equation of a circle:

$$(x - a)^2 + (y - b)^2 = r^2$$

where, (a, b) is the centre and r is the radius

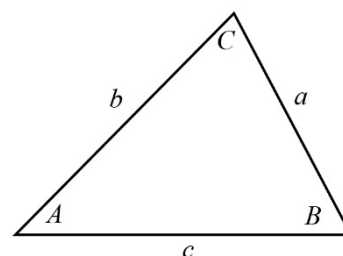
Trigonometric functions

Cosine and sine rules

For any triangle ABC with corresponding length of sides a, b, c

Cosine rule: $c^2 = a^2 + b^2 - 2ab \cos C$

Sine rule: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$



Area of Δ : $A = \frac{1}{2} ab \sin C$

$$A = \sqrt{s(s-a)(s-b)(s-c)} \text{ where } s = \frac{1}{2}(a+b+c)$$

Circular measure and radian measure

In a circle of radius r , for an arc subtending angle θ (radians) at the centre

Length of arc:	$\ell = r\theta$	Length of chord:	$l = 2r \sin \frac{1}{2}\theta$
Area of sector:	$A = \frac{1}{2}r^2\theta$	Area of segments:	$A = \frac{1}{2}r^2(\theta - \sin \theta)$

Trigonometric functions: (fundamentals)

$\sin(-\theta) = -\sin \theta$	$\cos(-\theta) = \cos \theta$	$\tan(-\theta) = -\tan \theta$
$\sin\left(\theta + \frac{\pi}{2}\right) = \cos \theta$	$\cos\left(\theta - \frac{\pi}{2}\right) = \sin \theta$	$\sin^2 \theta + \cos^2 \theta = 1$

Angle sum and difference identities

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

Counting and probability

Combinations

Number of combinations:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

(of r objects taken from a set of n distinct objects)

Binomial expansion:

$$(x + y)^n = x^n + \binom{n}{1}x^{n-1}y + \cdots + \binom{n}{r}x^{n-r}y^r + \cdots + y^n$$

Binomial coefficients:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n \times (n-1) \times \cdots \times (n-r+1)}{r \times (r-1) \times \cdots \times 2 \times 1}$$

Probability

Fundamentals of probability:

$$P(\text{complement of } A) = P(\bar{A}) = 1 - P(A)$$

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \text{ and } B) = P(A \cap B) = P(A)P(B|A)$$

$$= P(B)P(A|B)$$

Conditional probability:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \text{ for } P(A) \neq 0$$

Exponential functions

Index laws:

For $a, b > 0$ and m, n real,

$$a^m b^m = (ab)^m$$

$$a^m a^n = a^{m+n}$$

$$(a^m)^n = a^{mn}$$

$$a^{-m} = \frac{1}{a^m}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$a^0 = 1$$

For $a > 0$, m an integer and n a positive integer, $a^{\frac{m}{n}} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$

Arithmetic and geometric sequences and series

Arithmetic sequences

For initial term a and common difference d : $T_n = a + (n-1)d, n \geq 1$

$$T_{n+1} = T_n + d, \text{ where } T_1 = a$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

Geometric sequences

For initial term a and common ratio r :

$$T_{n+1} = rT_n, \text{ where } T_1 = a$$

$$T_n = ar^{n-1}, n \geq 1$$

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$S_\infty = \frac{a}{1-r}, |r| < 1$$

Introduction to differential calculus

Rates of change

Difference quotient:
$$\frac{\delta y}{\delta x} = \frac{f(x+h) - f(x)}{h}$$

Derivative (concept):
$$\frac{dy}{dx} = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Computation of derivatives:
$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

Anti-derivatives:
$$\text{If } f'(x) = ax^n \text{ then } f(x) = \frac{ax^{n+1}}{n+1} + c(\text{constant}), n \neq -1$$

Note: Any additional formulas identified by the examination writers as necessary will be included in the body of the particular question.